Nathaniel Powell

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Available for a 4-month co-op beginning January 2027

Education

University of Waterloo — Waterloo, ON

Bachelor of Science, Honours Mathematical Physics, Co-op — *Sep. 2025 – Present*

- Coursework and experience spanning applied mathematics, physics, programming, and computational problem solving.

Technical Skills

Product & Security: Requirements Gathering, User Feedback, Feature Prioritization, Authentication, Authorization, Role-Based Access Control (RBAC), Access Control
Languages & Development: Python, Java, JavaScript, SQL, React, Node.js, Supabase/PostgreSQL, REST APIs
Software Engineering: System Design, Data Modeling, Testing, Debugging, Performance Optimization, Object-Oriented Programming, Version Control
Tools: Git, GitHub, Figma, Microsoft Excel, PowerPoint

Experience

Quadmates.ca — Waterloo, ON

Founder & Developer — *2026 – Present*

- Led product development for a student-focused platform supporting approximately **50 early users**, gathering feedback and translating user needs into product and technical requirements.
- Prioritized feature improvements across authentication, student profiles, messaging, and study-partner matching based on **user feedback, product needs, and technical feasibility**.
- Designed and implemented **3 end-to-end workflows** across frontend, backend, API, authentication, and database layers using React, Node.js, JavaScript, and Supabase/PostgreSQL.
- Designed application architecture, database schemas, APIs, and data flows while applying authentication and access-control principles to user-facing functionality.
- Performed **performance optimization** across direct and group messaging workflows, reducing latency by **70%** and improving the user experience.
- Integrated **AI-assisted development workflows** into debugging, code generation, refactoring, and feature implementation to accelerate product iteration and technical problem-solving.

UW Orbital — Waterloo, ON

Business & Finance Team Member — *Jan. 2026 – Present*

- Developing a **Python automation workflow** to streamline expense-data processing and reduce manual updates to financial records.
- Built and maintain an **Excel-based structured dataset** to centralize spending information, organize financial records, and support project budget tracking.
- Collaborate across a **406-member multidisciplinary engineering organization**, working with technical and business teams to support project requirements and organizational priorities.
- Conducted external stakeholder outreach to a company of approximately **50 employees** to coordinate potential fabrication materials and engineering support.

Projects

Staffflow | *Staff Management Application*

- Developed a staff-management application with distinct **manager, technical, employee, and administrator roles**, applying role-based access control to separate user permissions and application functionality.
- Designed authentication and authorization workflows to restrict access to features and administrative functions based on assigned user roles.
- Built role-specific workflows and interfaces around the needs and permissions of different user groups, applying **product-security and access-control principles**.